

Analysis

Parameter Selection:

All simulation parameters were selected using conservative values taken from real world drone specifications. Choosing values must account for payload, safety margins, and operating variability, while ensuring that all timing calculations remain realistic and consistent across different drone models.

Travel Time:

The travel time table shows that acceleration and deceleration add a fixed time to every mission. For shorter distances, this fixed time makes up a large portion of total travel time, which supports modeling drone motion using acceleration, cruise, and deceleration phases instead of assuming constant speed.

Mission Time:

The mission time breakdown shows that total mission time increases with fire severity because higher severities require more water or foam. For high severity fires, the drop time is the largest part of the mission, meaning overall response time is more limited by delivery than by travel.

Reference Drones:

The reference drone table shows that maximum speed and spray rate vary across different commercial drones. Using a conservative choice for cruise speed and spray rate ensures a realistic simulation under full payload conditions.